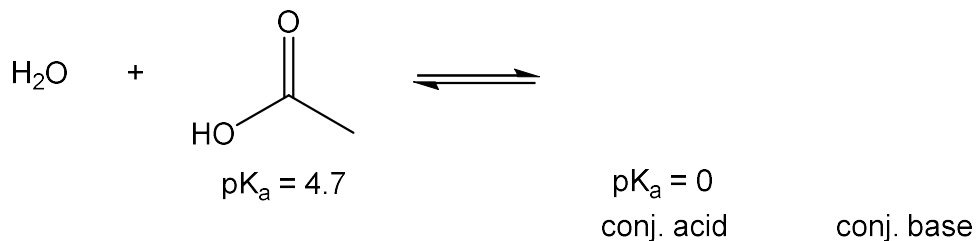
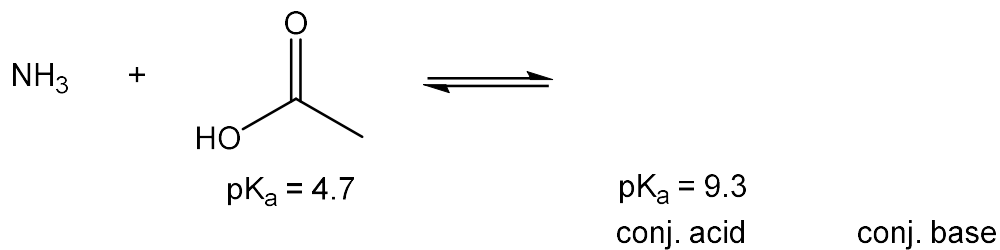
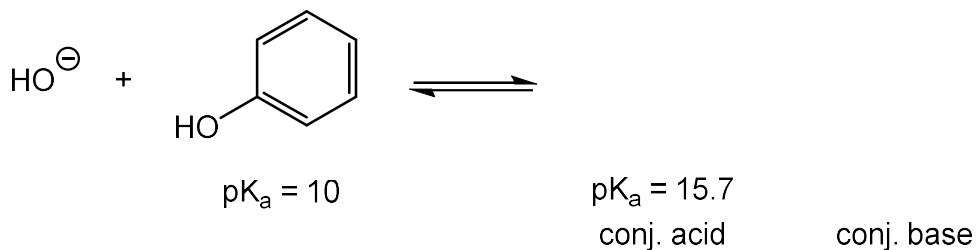


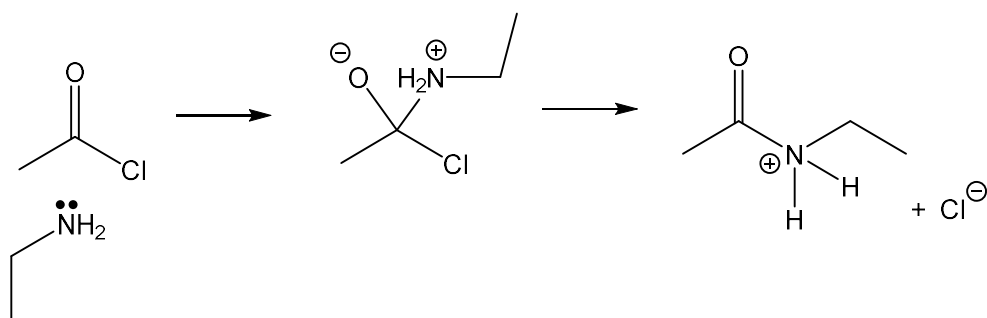
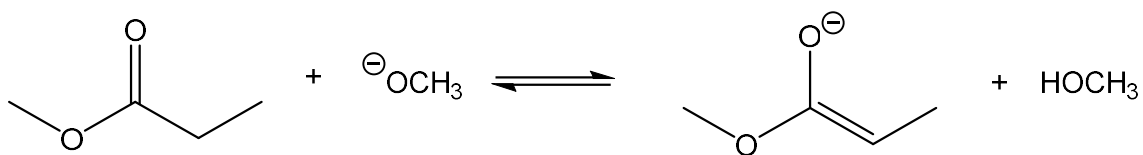
Bonding, Acid/Base Worksheet

1. Draw a Lewis structure for the nitrite ion (NO_2^{1-}). Considering that a resonance form can be drawn, what is the hybridization of the O atom labeled with a negative charge? Draw separate figures showing the sigma and pi bonding present in NO_2^{1-} .

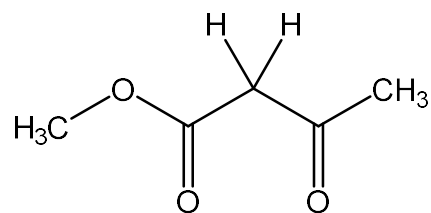
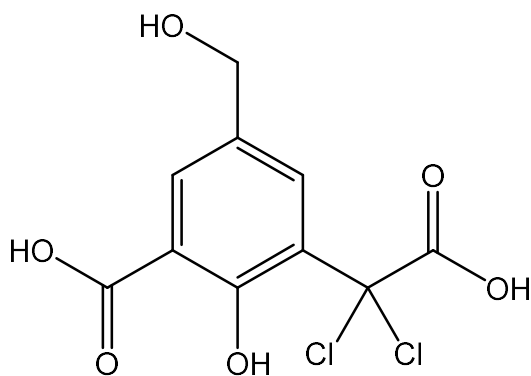
2. Draw the structures of the conjugate acid/base formed in the following reactions. Is the reaction product favored (is the initial base "strong enough")?



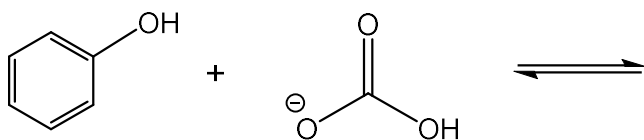
3. Fill in the missing arrows for the following reactions (moving from left to right).



4. Rank the relative acid strengths of the bold H atoms in each molecule (1 being the most acidic).



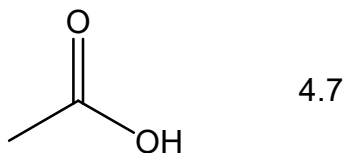
5. Complete the following acid-base reaction. Match the pKa values given with the appropriate acid on either side of the equilibrium. Determine the equilibrium constant (K_{eq}) and from that value, decide which side of the equilibrium is favored.



pKa
6.4
10

6. Draw the structure for each of the following compounds at pH = 6.

pKa pH = 6



ammonium 9.4